Shivalika Karan Bora

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Profile

MSc Data Science and Analytics student at Cardiff University (graduating May 2027, available full time from August 2027) with 2+ years of software engineering experience at Tata Consultancy Services. I build machine learning models and data pipelines, from forecasting energy use across 20M+ records to putting models into working web apps. Strong in Python, SQL and APIs, with experience delivering reliable data systems for a national-scale government service.

Education

MSc Data Science and Analytics, Cardiff University | Jan 2026 – May 2027 (expected)

Team projects in machine learning, time-series forecasting and full-stack data applications. Learning R through coursework.

B.Tech in Computer Science and Engineering, SRM Institute of Science and Technology | 2019 – 2023 | CGPA 9.04/10

Professional Experience

Software Engineer, Tata Consultancy Services | 2023 – 2025

- Built and maintained 50+ REST APIs and data integration services for a national government system used across India.
- Wrote SQL (MySQL) queries, data validation rules and batch processing jobs to keep high-volume data accurate and ready for reporting.
- Automated MIS reporting and batch workflows, cutting manual effort by 40%.
- Implemented Redis caching to improve the speed and reliability of high-traffic services.
- Tested features, resolved production issues and supported releases in an Agile team. Recognised with 4 performance awards.

Technical Intern, Tata Power-DDL | 2022 | Built a Python image analysis application (filtering, edge detection, histograms) and supported system troubleshooting.

Software Development Intern, HighRadius | 2021 | Contributed to a full-stack web project across frontend and backend in an Agile team.

Projects

Building Energy Prediction (ASHRAE) | Python, Pandas, LightGBM, XGBoost, CatBoost | Team of six

- Cleaned and merged 20M+ hourly meter readings from about 1,450 buildings with building and weather data, engineering lag, rolling-window and calendar features.
- Built and tuned the LightGBM model (RMSLE 0.523). The team's ensemble cut error by about 61% against the median baseline.

WasteWise AI | React, FastAPI, PostgreSQL, Docker | Team project

- Built the React frontend, login flow, protected image-classification API, prediction history and bin feedback for an app that sorts household waste into 30 categories.
- Integrated a four-model CNN ensemble (MobileNetV2, EfficientNet-B0, ResNet50, DenseNet121) into the production endpoint, storing results in PostgreSQL.

AI Jobs Market Dashboard | Python, Pandas, Streamlit, Plotly | Solo project

- Cleaned and merged two job-market datasets into KPIs, six interactive chart types and auto-generated insights on AI hiring trends.

Technical Skills

- **Programming and data:** Python, SQL, Pandas, NumPy, R (learning), Java
- **Machine learning:** scikit-learn, LightGBM, XGBoost, CatBoost, time-series forecasting, model evaluation
- **Data engineering:** data pipelines, ETL, data validation, data quality, batch processing
- **Databases:** MySQL, PostgreSQL, Redis
- **Engineering:** REST APIs, FastAPI, Spring Boot, React, Docker, Git/GitLab, Agile
- **Visualisation:** Streamlit, Plotly, dashboards

Leadership

Residence Life Assistant, Cardiff University | Jan 2026 – Jun 2027

- Organise community events for residents, including weekly drop-ins and Diwali, Holi and craft events.
- First point of contact for student queries, signposting residents to the right support.